



2008

TRIAL
HIGHER SCHOOL CERTIFICATE

GIRRAWEEEN HIGH SCHOOL

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board – approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1 – 10
- All questions are of equal value

Girraween High School_Mathematics_Trial HSC_2008.

Total marks – 120

Attempt Questions 1 – 10

All questions are of equal value

Question 1 (12 marks).

Marks

- | | | |
|-----|---|---|
| (a) | Evaluate $\frac{\pi+2}{\pi-2}$ correct to one decimal place. | 2 |
| (b) | Solve $2x+1 \leq 7$ and graph the solution on a number line. | 2 |
| (c) | Rationalise the denominator of $\frac{1}{\sqrt{6}-2}$. | 2 |
| (d) | Find the limiting sum of the geometric series
$9 - 3 + 1 - \dots$ | 2 |
| (e) | Factorise $6x^2 - x - 2$. | 2 |
| (f) | At Octopus Fones annual sale, all mobile phones are discounted by 40%. Mia paid \$630 for a mobile phone at the sale. What was the original price of the phone? | 2 |

Marks

Question 2 (12 marks). Start on a SEPARATE page.

(a) Differentiate with respect to x :

(i) $\frac{1}{x}$.

1

(ii) $\frac{3x}{e^x - 1}$.

2

(iii) $(1 + \cos x)^8$.

2

(b) (i) Find $\int 2\sec^2 3x dx$.

2

(ii) Evaluate $\int_0^1 \frac{2x}{x^2 + 1} dx$.

2

(Leave answer in exact form).

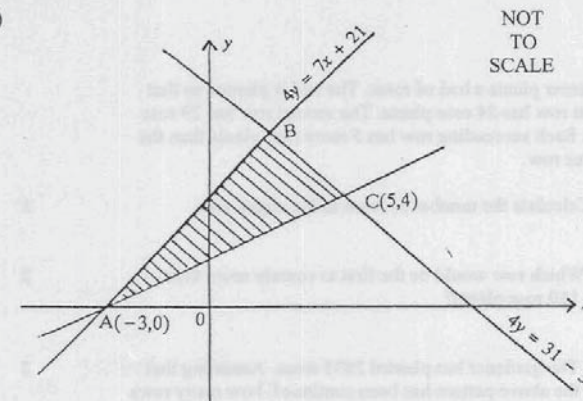
(c) Find the equation of the normal to the curve $y = e^{4x} - 1$ at the point on the curve where $x = 0$.

3

Marks

Question 3 (12 marks). Start on a SEPARATE page.

(a)



In the diagram, the lines $4y = 7x + 21$ and $4y = 31 - 3x$ intersect at the point B. A and C are the points $(-3, 0)$ and $(5, 4)$ respectively.

(i) Calculate the gradient of AC.

1

(ii) Show that the line AC has equation $x - 2y + 3 = 0$.

1

(iii) Show that B has coordinates $(1, 7)$.

1

(iv) Show that the perpendicular distance from B to the line AC is $2\sqrt{5}$ units.

2

(v) Find the exact length of the interval AC. (Express answer as a simplified surd).

1

(vi) Find the area of triangle ABC.

1

Question 3 continues on page 5

Question 3 (continued)

Marks

- (b) A gardener plants a bed of roses. The bed is planted so that the first row has 24 rose plants. The second row has 29 rose plants. Each succeeding row has 5 more rose plants than the previous row.

- | | |
|--|---|
| (i) Calculate the number of roses in the eighth row. | 1 |
| (ii) Which row would be the first to contain more than 150 rose plants? | 2 |
| (iii) The gardener has planted 2895 roses. Assuming that the above pattern has been continued, how many rows were planted? | 2 |

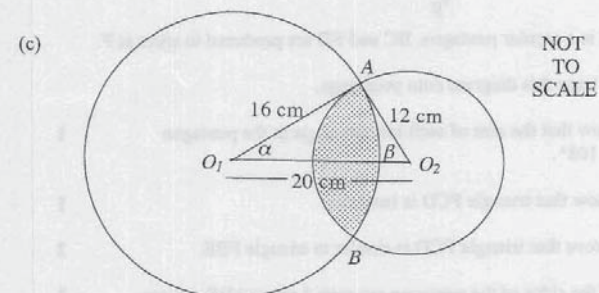
End of Question 3

Please turn over

Marks

Question 4 (12 marks). Start on a SEPARATE page.

- | | |
|---|---|
| (a) The sides of a triangle are 7 cm, 5 cm and 4 cm. Find the size of the angle opposite the largest side. (Give answer correct to the nearest minute). | 2 |
| (b) A fair die is rolled twice. Find the probability that: | |
| (i) the second score is greater than the first score. | 2 |
| (ii) the total of the two scores is 7 or 11. | 1 |

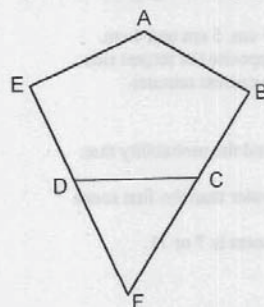


A circle with centre at O_1 , and radius 16 cm intersects with another circle with centre at O_2 and radius 12 cm. Their points of intersection are A and B and the distance between their centres, O_1O_2 , is 20 cm. The $\angle AO_1O_2 = \alpha$ and $\angle AO_2O_1 = \beta$.

- | | |
|--|---|
| (i) Show that triangle O_1AO_2 is a right-angled triangle. | 1 |
| (ii) Find the area of the quadrilateral AO_2BO_1 . | 1 |
| (iii) Find the size of the angles α and β . | 2 |
| (iv) Find the shaded area enclosed by these circles. (Give your answer correct to the nearest cm^2). | 3 |

Marks

Question 5 (12 marks). Start on a SEPARATE page.

NOT
TO
SCALE

ABCDE is a regular pentagon. BC and ED are produced to meet at F.

Copy or trace this diagram onto your page.

- | | |
|--|---|
| (i) Show that the size of each interior angle in the pentagon is 108° . | 1 |
| (ii) Show that triangle FCD is isosceles. | 1 |
| (iii) Prove that triangle FCD is similar to triangle FBE. | 2 |
| (iv) If the sides of the pentagon are each 5 cm and $BE = 8$ cm, determine the length of CF. | 2 |
- (b) If α and β are the roots of the quadratic equation $3x^2 + 4x + 7 = 0$, find the values of:
- | | |
|---|---|
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\alpha^2 + \beta^2$ | 1 |
| (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 1 |

Question 5 continues on page 8

Question 5 (continued)

Marks

- (c) A particle moves in a straight line so that its acceleration, a m/s^2 , at time t seconds is given by $a = 3(4+t)^2$. Initially the particle is moving with a velocity of $64 m/s$. Find the velocity of the particle as a function of time. 2

Question 6 (12 marks). Start on a SEPARATE page.

- (a) Solve the following equation for x : 2
- $$e^{2x} - e^x - 6 = 0.$$
- (b) Let $f(x) = \frac{4x^3 - x^4}{9}$.
- | | |
|---|---|
| (i) Find the coordinates of the points where the curve crosses the axes. | 2 |
| (ii) Find any stationary points, and determine their nature. | 3 |
| (iii) Find any points of inflexion. | 3 |
| (iv) Sketch the graph of $y = f(x)$, indicating clearly the intercepts, stationary points and points of inflexion. | 2 |

End of Question 6

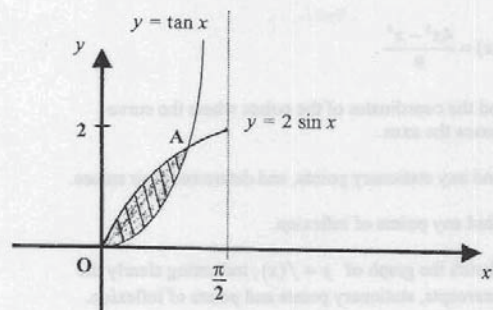
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Marks

Question 7 (12 marks). Start on a SEPARATE page.

- (a) A parabola whose equation is $y = kx^2$, where k is a constant, has the line $y = -6x + 3$ as a tangent.
- By equating the two given equations, find a quadratic equation in terms of x and k . 1
 - By using the discriminant of the quadratic equation found, find the value of k . 2
 - Find the coordinates of the point of contact between the tangent and the parabola. 2
 - Sketch the parabola and the tangent line, showing the coordinates of the point of contact and where the tangent line cuts the x - and y -axes. 2

(b)



The diagram shows the graphs of $y = 2 \sin x$ and $y = \tan x$ for $0 \leq x \leq \frac{\pi}{2}$. A is the point of intersection of the two graphs.

- Find the coordinates of point A . 2
- Show that $\frac{d}{dx}[(\ln \cos x)] = -\tan x$ 1
- Find the area of the shaded region in the diagram. 2

Marks

Question 8 (12 marks). Start on a SEPARATE page.

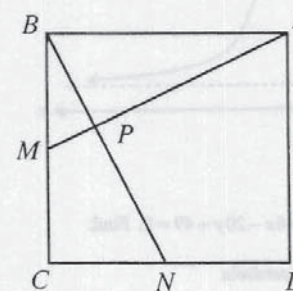
- (a) A population of bacteria in a medium are growing at a rate proportional to the current population. The population obeys the model

$$P = P_0 e^{kt}$$

where P_0 is the population of bacteria at noon on 1 August and t is measured in hours. When $t = 6$ the population has grown from 900 000 to 1.4 million.

- Show that $\frac{dP}{dt} = kP$ 1
- What is the value of k ? 2
- What will the population be when $t = 10$? 1
- When will the population reach 3 million? 1

(b)



$ABCD$ is a square. M and N are the midpoints of BC and CD respectively.

- Prove triangles ABM and BCN are congruent. 4
- Prove that AM and BN are perpendicular. 3

Marks

Question 9 (12 marks). Start on a SEPARATE page.

- (a) The table shows the values of a function
- $f(x)$
- for four values of
- x
- .

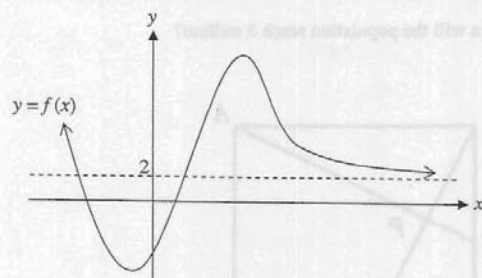
x	2	3	4	5
$f(x)$	0.693	1.099	1.386	1.609

Use the trapezoidal rule with these four values to find an approximation to $\int_2^5 f(x) dx$.

2

- (b) The diagram below shows a sketch of the curve $y = f(x)$. Copy or trace the diagram on your page and use it to draw a sketch of the gradient function $y = f'(x)$.

2



- (c) A parabola has equation $x^2 + 6x - 20y + 49 = 0$. Find:

- the focal length of this parabola
- the coordinates of the vertex
- the coordinates of the focus
- the equation of the directrix

2

1

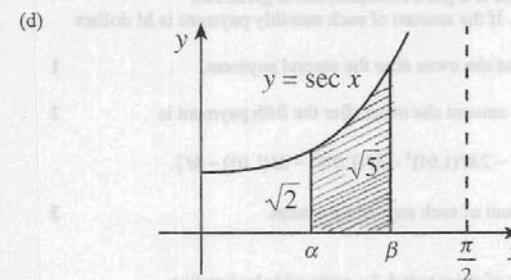
1

1

Question 9 continues on page 12

Marks

Question 9 (continued)



The shaded region in the diagram is bounded by the curve $y = \sec x$, the x -axis and the lines $x = \alpha$ and $x = \beta$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.

3

End of Question 9

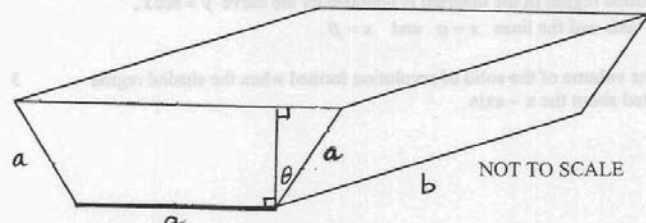
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Question 10 (12 marks). Start on a SEPARATE page.

- (a) Arina borrows \$20000 from City Credit at 12% p.a. interest. She pays it back at regular monthly intervals over four years. However, because she is a good customer, she is given two months interest free. If the amount of each monthly payment is M dollars

- (i) Find the amount she owes after the second payment. 1
- (ii) Show that the amount she owes after the fifth payment is 2
 $\$[20000(1.01)^3 - 2M(1.01)^3 - M(1.01)^2 - M(1.01) - M].$
- (iii) Find the amount of each monthly payment. 3

- (b) A gutter is made out of sheet metal $3a$ units wide by bending it as shown in the diagram. The length of the gutter is b units.



- (i) Show that the volume of the gutter is given by 2
 $V = a^2 b \cos \theta (1 + \sin \theta).$
- (ii) Show that $\frac{dV}{d\theta} = a^2 b (1 - 2 \sin^2 \theta - \sin \theta).$ 2
- (iii) Determine the value of θ , in degrees, so that the gutter has maximum volume. 2

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 1

(a) $\frac{\pi+2}{\pi-2} = 4.503876788$
 $= 4.5$ (1 dec. pl.) (2)

(b) $2x+1 \leq 7$
 $2x \leq 6$
 $\therefore x \leq 3$
 (2)

(c) $\frac{1}{\sqrt{6}-2} = \frac{1}{\sqrt{6}-2} \times \frac{\sqrt{6}+2}{\sqrt{6}+2}$
 $= \frac{\sqrt{6}+2}{6-4}$
 $= \frac{\sqrt{6}+2}{2}$ (2)

(d) $9-3+1 = \dots$
 $a=9, r = -\frac{3}{9} = -\frac{1}{3}$
 $S_{\infty} = \frac{a}{1-r}$
 $= \frac{9}{1+\frac{1}{3}}$
 $= 6\frac{3}{4}$ (2)

(e) $6x^2 - x - 2$
 $= (2x+1)(3x-2)$ (2)

(f) $60\% = \$630$
 $1\% = \frac{\$630}{60}$
 $\therefore 100\% = \frac{\$630}{60} \times 100$
 $= \$1050$ (2)

Question 2

(a) (i) $\frac{d}{dx}\left(\frac{1}{x}\right)$
 $= -1x^{-2}$
 $= -\frac{1}{x^2}$ (1)

(ii) $\frac{d}{dx}\left(\frac{3x}{e^x-1}\right)$
 $= \frac{(e^x-1) \cdot 3 - 3x(e^x)}{(e^x-1)^2}$
 $= \frac{3e^x - 3 - 3xe^x}{(e^x-1)^2}$
 $= \frac{3(e^x - 1 - xe^x)}{(e^x-1)^2}$ (2)

(iii) $\frac{d}{dx}[(1+\cos x)^8]$
 $= 8(1+\cos x)^7 \times \frac{d}{dx}(\cos x)$
 $= -8\sin x(1+\cos x)^7$ (2)

(b) (i) $\int 2\sec^2 3x \, dx$
 $= \frac{2\tan 3x}{3} + c$ (2)

(ii) $\int \frac{2x}{x^2+1} \, dx$
 $= [\ln(x^2+1)]_0^1$
 $= \ln 2 - \ln 1$
 $= \ln 2$ (2)

(c) $y = e^{4x} - 1$
 $\therefore \frac{dy}{dx} = 4e^{4x}$

when $x=0, y = e^0 - 1 = 0$
 $\therefore x=0, \frac{dy}{dx} = 4e^0 = 4$

$\therefore$ gradient of normal is $-\frac{1}{4}$
 $\therefore$ equation of normal is
 $y-0 = -\frac{1}{4}(x-0)$
 $\therefore y = -\frac{1}{4}x$
 $\text{or } x+4y=0$ (3)

Question 3

(a) (i) $m_{AC} = \frac{4-0}{5+3}$
 $= \frac{4}{8}$
 $= \frac{1}{2}$ (1)

(ii) using $(-3,0)$ and $m = \frac{1}{2}$
 $\therefore$ equation of AC is
 $y-0 = \frac{1}{2}(x+3)$
 $\therefore 2y = x+3$
 $\therefore x-2y+3=0$ (1)

(iii) $4y = 7x+21 \dots (1)$
 $4y = 31-3x \dots (2)$
 $\therefore 7x+21 = 31-3x$
 $7x+3x = 31-21$
 $10x = 10$
 $\therefore x = 1$

using (1), $4y = 7(1)+21$
 $\therefore 4y = 28$

$\therefore y = 7$
 $\therefore B(1,7)$ (1)

(iv) $d = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$
 $\text{using } B(1,7) \text{ \& } x-2y+3=0$
 $\therefore d = \frac{|(1)(1)+(-2)(7)+3|}{\sqrt{1^2+(-2)^2}}$
 $= \frac{|-10|}{\sqrt{5}}$

$= \frac{10}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$
 $= \frac{10\sqrt{5}}{5}$
 $= 2\sqrt{5} \text{ units}$ (2)

(v) $AC = \sqrt{(5+3)^2 + (4-0)^2}$
 $= \sqrt{64+16}$
 $= \sqrt{80}$
 $= \sqrt{16 \times 5}$
 $= 4\sqrt{5} \text{ units}$ (1)

(vi) Area $\triangle ABC$
 $= \frac{1}{2} \times AC \times d$
 $= \frac{1}{2} \times 4\sqrt{5} \times 2\sqrt{5}$
 $= 20 \text{ unit}^2$ (1)

(b) 24, 29, 34,
A.P. with $a=24$, $d=5$

$$(i) T_n = a + (n-1)d$$

$$\therefore T_8 = 24 + 7 \times 5$$

$$= 59. \quad (1)$$

$$(ii) 24 + (n-1)5 > 150$$

$$24 + 5n - 5 > 150$$

$$19 + 5n > 150$$

$$5n > 131$$

$$\therefore n > 26\frac{1}{5}$$

$$\therefore n = 27. \quad (2)$$

$$(iii) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\therefore 2895 = \frac{n}{2} [2 \times 24 + (n-1)5]$$

$$5790 = n [48 + (n-1)5]$$

$$5790 = n (43 + 5n)$$

$$\therefore 5n^2 + 43n - 5790 = 0$$

$$\therefore n = \frac{-43 \pm \sqrt{43^2 + 4 \times 5 \times 5790}}{2 \times 5}$$

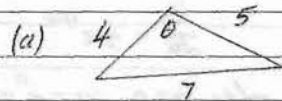
$$= \frac{-43 \pm \sqrt{117649}}{10}$$

$$= \frac{-43 \pm 343}{10}$$

$$= 30 \text{ or } -38.6$$

$$\therefore n = 30 \text{ since } n > 0 \quad (2)$$

Question 4



Using cosine rule,

$$\cos \theta = \frac{4^2 + 5^2 - 7^2}{2 \times 4 \times 5}$$

$$= \frac{8}{40}$$

$$\therefore \theta = 101^\circ 32'. \quad (2)$$

(b) (i)

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

P(2nd score > 1st score) (1)

$$= \frac{15}{36}$$

$$= \frac{5}{12}. \quad (1)$$

(ii) P(total of 2 scores is 7 or 11,

$$= \frac{8}{36}$$

$$= \frac{2}{9}. \quad (1)$$

(c) (i) $16^2 = 256$
 $12^2 = 144$
 $20^2 = 400.$

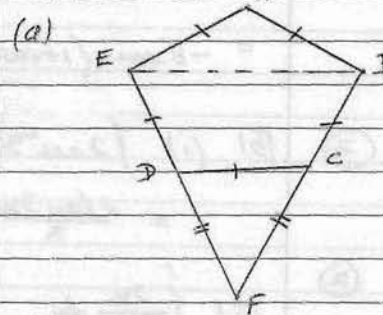
Since $20^2 = 16^2 + 12^2$
 Δ is right-angled at A (1)

(ii) Area of $\Delta O_2 B O_1$
 $= 2 \times \left(\frac{1}{2} \times 16 \times 12 \right)$
 $= 192 \text{ cm}^2 \quad (1)$

(iii) $\sin \alpha = \frac{12}{20}$
 $\therefore \alpha = 36^\circ 52'$
 $\therefore \beta = 90^\circ - 36^\circ 52'$
 $= 53^\circ 8' \quad (2)$

(iv) Shaded area
 $= \frac{1}{2} \times 16^2 (2 \times 0.644 - \sin 2 \times 0.644)$
 $+ \frac{1}{2} \times 12^2 (2 \times 0.927 - \sin 2 \times 0.927)$
 $= 106 \text{ cm}^2 \text{ (nearest cm}^2 \text{)} \quad (3)$

Question 5



(i)
 $\angle \text{sum of polygon} = (n-2) \times 180^\circ$
 $\therefore \angle \text{sum of pentagon} = (5-2) \times 180^\circ$
 $= 540^\circ$
 $\therefore \text{each interior } \angle = \frac{540^\circ}{5} = 108^\circ \quad (1)$

(ii) $\angle EDC + \angle CDF = 180^\circ$ (str. $\angle$)
 $\therefore 108^\circ + \angle CDF = 180^\circ$
 $\therefore \angle CDF = 72^\circ$

Similarly, $\angle DCF = 72^\circ$
 $\therefore \Delta FCD$ is isosceles (1)

(iii) $FC = FD$ (isos. Δ CDF,
 $CB = DE$ (sides of regular pentagon))

$$\therefore FB = FE$$

$$\therefore \frac{FC}{FB} = \frac{FD}{FE}$$

$\angle F$ is common

$$\therefore \Delta FCD \parallel \Delta FBE \quad (2)$$

(iv) $\frac{FC}{FB} = \frac{DC}{EB}$ (matching sides of sim. Δ s proportion)

Let $FC = x$,

$$\therefore \frac{x}{x+5} = \frac{5}{8}$$

$$\therefore 8x = 5(x+5)$$

$$8x = 5x + 25$$

$$\therefore 3x = 25 \quad (2)$$

$$\therefore x = 8\frac{1}{3} \text{ cm}$$

(b) (i) $3x^2 + 4x + 7 = 0$

$$\alpha + \beta = \frac{-4}{3} \quad (1)$$

(ii) $\alpha\beta = \frac{7}{3} \quad (1)$

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(-\frac{4}{3}\right)^2 - 2 \times \frac{1}{3}$$

$$= \frac{16}{9} - \frac{14}{3}$$

$$= -2\frac{8}{9} \quad (1)$$

$$(iv) \frac{1}{\alpha} + \frac{1}{\beta}$$

$$= \frac{\alpha + \beta}{\alpha\beta}$$

$$= -\frac{4}{3} \div \frac{7}{3}$$

$$= -\frac{4}{7} \quad (1)$$

$$(c) \quad a = 3(4+t)^2$$

$$v = \int 3(4+t)^2 dt$$

$$= \frac{3(4+t)^3}{3} + C$$

$$= (4+t)^3 + C$$

when $t=0$, $v=64$

$$64 = C + 4^3$$

$$\therefore C=0$$

$$\therefore v = (4+t)^3 \quad (2)$$

Question 6

$$(a) \quad e^{2x} - e^x - 6 = 0$$

Let $t = e^x$

$$\therefore t^2 - t - 6 = 0$$

$$(t-3)(t+2) = 0$$

$$\therefore t = 3 \text{ or } t = -2$$

$$e^x = 3 \text{ or } e^x = -2$$

(no solution)

$$\therefore x \ln e = \ln 3$$

$$\therefore x = \ln 3 \quad (2)$$

$$(b) \quad f(x) = \frac{4x^3 - x^4}{9}$$

(i) Curve cuts x -axis when

$$f(x) = 0$$

$$\therefore \frac{4x^3 - x^4}{9} = 0$$

$$4x^3 - x^4 = 0 \text{ since } 9 \neq 0$$

$$x^3(4-x) = 0 \quad (2)$$

$$\therefore x = 0 \text{ or } x = 4$$

$\therefore$ coords. are $(0,0)$ & $(4,0)$

(ii) Stat. pts. when $f'(x) = 0$

$$f'(x) = \frac{1}{9}(12x^2 - 4x^3)$$

$$= 0$$

$$\therefore 12x^2 - 4x^3 = 0$$

$$4x^2(3-x) = 0$$

$$\therefore x = 0 \text{ or } x = 3$$

When $x=0$, $y=0$

$$\therefore x=3, y = \frac{4 \times 3^3 - 3^4}{9} = 3$$

stat. pts. at $(0,0)$ & $(3,3)$

Test $(0,0)$:

When $x = -0.1$, $f'(x) = 0.014 > 0$

" $x = 0.1$, $f'(x) = 0.013 > 0$

$\therefore (0,0)$ is a horizontal P.O.I

Test $(3,3)$:

When $x = 2.9$, $f'(x) = 0.37 > 0$

" $x = 3.1$, $f'(x) = -0.43 < 0$

$\therefore$ max. & pt. at $(3,3)$ (3)

(iii) Possible pts. of inflexion when $f''(x) = 0$

$$f''(x) = \frac{1}{9}(24x - 12x^2)$$

$$= 0$$

$$\therefore 24x - 12x^2 = 0$$

$$12x(2-x) = 0$$

$$\therefore x = 0 \text{ or } x = 2$$

when $x=0$, $y=0$

" $x=2$, $y = \frac{4 \times 2^3 - 2^4}{9} = \frac{16}{9}$

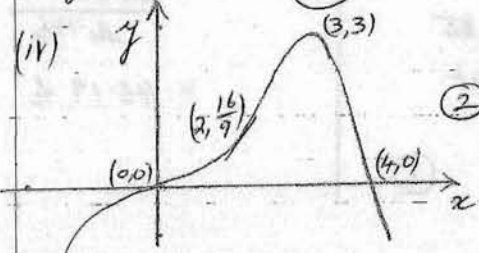
Now $(0,0)$ is a horizontal pt. of inflexion from (ii) above.

Test $(2, \frac{16}{9})$:

When $x = 1.9$, $f''(x) = 0.25 > 0$

" $x = 2.1$, $f''(x) = -0.28 < 0$

$\therefore$ since change in concavity occurs $(2, \frac{16}{9})$ is a point of inflexion. (3)



Question 7

(a) (i) $kx^2 = -6x + 3$

$$\therefore kx^2 + 6x - 3 = 0 \quad (1)$$

(ii) $\Delta = 0$,

$$\therefore 6^2 - 4k(-3) = 0$$

$$36 + 12k = 0$$

$$\therefore k = -3 \quad (2)$$

(iii) $-3x^2 + 6x - 3 = 0$

$$\therefore 3x^2 - 6x + 3 = 0$$

$$3(x^2 - 2x + 1) = 0$$

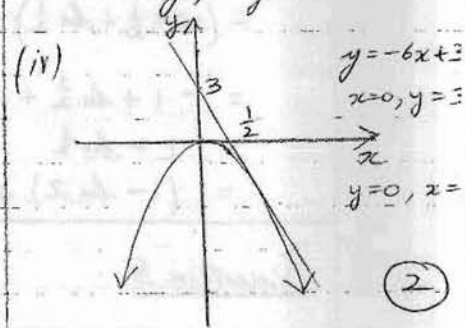
$$3(x-1)^2 = 0$$

$$\therefore x = 1 \quad (2)$$

using $y = -3x^2$

$$= -3 \times 1^2 = -3$$

$\therefore$ coords. of pt. of contact $(1, -3)$



(b) (i) $2 \sin x = \tan x$

$$= \frac{\sin x}{\cos x} \quad (\cos x \neq 0)$$

$$\therefore 2 \sin x \cos x = \sin x$$

$$\therefore 2 \sin x \cos x - \sin x = 0$$

$$\begin{aligned} \therefore \sin x (2 \cos x - 1) &= 0 \\ \therefore \sin x &= 0 \text{ or } 2 \cos x - 1 = 0 \\ \therefore x &= 0 \text{ or } \cos x = \frac{1}{2} \\ \therefore x &= \pi/3 \end{aligned}$$

$$\begin{aligned} \text{when } x &= \pi/3, \\ y &= \tan \pi/3 = \sqrt{3} \\ \therefore A &(\frac{\pi}{3}, \sqrt{3}) \quad (2) \end{aligned}$$

$$\begin{aligned} (ii) \frac{d}{dx} [\ln \cos x] &= \frac{-\sin x}{\cos x} \\ &= -\tan x \quad (1) \end{aligned}$$

$$\begin{aligned} (iii) A &= \int_0^{\pi/3} (2 \sin x - \tan x) dx \\ &= [-2 \cos x + \ln(\cos x)]_0^{\pi/3} \\ &= (-2 \times \frac{1}{2} + \ln \frac{1}{2}) - (-2 \times 1 + \ln 1) \\ &= -1 + \ln \frac{1}{2} + 2 \\ &= 1 + \ln \frac{1}{2} \quad (2) \\ &= (1 - \ln 2) \text{ units}^2 \end{aligned}$$

Question 8

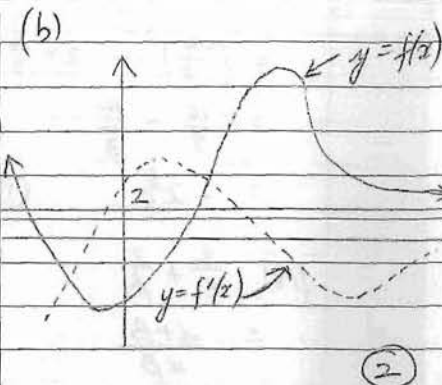
$$\begin{aligned} (a) (i) P &= P_0 e^{kt} \\ \therefore \frac{dP}{dt} &= P_0 \times k e^{kt} \\ &= k \times P_0 e^{kt} \\ &= kP \quad (1) \end{aligned}$$

$$\begin{aligned} (ii) 1400000 &= 900000 e^{6k} \\ \frac{14}{9} &= e^{6k} \\ \therefore \ln \frac{14}{9} &= 6k \\ \therefore k &= \frac{1}{6} \ln \frac{14}{9} \quad (2) \\ \therefore k &= 0.0736 \text{ (4 d.p.)} \end{aligned}$$

$$\begin{aligned} (iii) \text{When } t &= 10, \\ P &= 900000 e^{10k} \\ \text{where } k &= \frac{1}{6} \ln \frac{14}{9} \\ &= 1879540.637 \quad (1) \\ &= 1879540 \text{ (nearest whole no.)} \end{aligned}$$

$$\begin{aligned} (iv) 3000000 &= 900000 e^{kt} \\ \text{where } k &= \frac{1}{6} \ln \frac{14}{9} \\ \therefore \frac{30}{9} &= e^{kt} \\ \therefore \ln \frac{30}{9} &= kt \\ \therefore t &= \frac{1}{k} \ln \frac{30}{9} \\ &= \ln \frac{30}{9} \div \frac{1}{6} \ln \frac{14}{9} \\ &= \frac{\ln 30 \times 6}{\ln 14/9} \\ &= 46.19 \text{ h} \quad (1) \end{aligned}$$

(i) In Δs ABM and BCN
 $AB = BC$ (sides of a square)
 $BM = CN$ (M, N are mid-point of equal sides)
 $\angle ABM = \angle BCN = 90^\circ$ (ls of a square)
 $\therefore \Delta ABM \equiv \Delta BCN$ (SAS)



(ii) $\angle CBN = \angle BAM$ (matching ls of congruent Δs).
 Let $\angle ABP = x^\circ$, $\angle BAM = y^\circ$
 $\therefore \angle CBN = y^\circ$
 $\therefore x + y = 90^\circ$ ($\angle ABC$)
 $\therefore \angle ABP = 90^\circ$ ($\angle$ sum of ΔABP)
 $\therefore AM \perp BN$. (3)

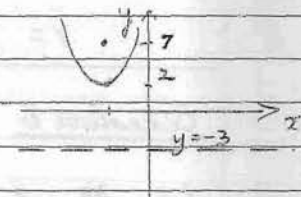
Question 9

$$\begin{aligned} (a) \int_2^5 f(x) dx &= \frac{1}{2} [0.693 + 2(1.099 + 1.386) + 1.609] \\ &= \frac{1}{2} \times 7.272 \\ &= 3.636. \quad (2) \end{aligned}$$

(c) (i) $x^2 + 6x = 20y - 49$
 $x^2 + 6x + 3^2 = 20y - 49 + 3^2$
 $(x+3)^2 = 20(y-2)$
 of form $(x-h)^2 = 4a(y-k)$
 $4a = 20$
 $\therefore a = 5$ (2)

(ii) Vertex (h, k)
 $\therefore V(-3, 2)$ (1)

(iii) Focus (-3, 7) (1)



(iv) Equation of directrix is $y = -3$. (1)

Question 10

(a) (i) $n = 48$, $r = 1\%$ p. month,

$$A_{48} = 0, R = 1.01.$$

$$A_1 = 20000 - M$$

$$A_2 = (20000 - M) - M$$

$$= 20000 - 2M.$$

(ii) $A_3 = (20000 - 2M) \cdot 1.01 - M$

$$= 20000(1.01) - 2M(1.01) - M$$

$$A_4 = A_3(1.01) - M$$

$$= [20000(1.01) - 2M(1.01) - M] \cdot 1.01 - M$$

$$- M$$

$$= 20000(1.01)^2 - 2M(1.01)^2 - M$$

$$- M(1.01) - M$$

$$A_5 = A_4(1.01) - M$$

$$= [20000(1.01) - 2M(1.01) - M(1.01) - M] \cdot 1.01 - M$$

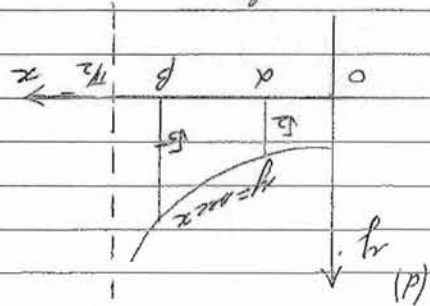
$$- M$$

$$- M$$

$$= 20000(1.01)^3 - 2M(1.01)^3 - M(1.01)^2 - M(1.01) - M$$

as required.

(2)



$$V = \pi \int_0^{\pi/2} \arctan x \, dx$$

$$= \pi [\tan x]_0^{\pi/2}$$

$$= \pi (\tan \beta - \tan \alpha)$$

Now $\sec \alpha = \sqrt{2}$ & $\sec \beta = \sqrt{5}$

$$\therefore \sec^2 \alpha = 2 \quad \sec^2 \beta = 5$$

$$\therefore 2 - \tan^2 \alpha = 1 \quad 5 - \tan^2 \beta = 1$$

$$\tan^2 \alpha = 1 \quad \tan^2 \beta = 4$$

$$\therefore \tan \alpha = \pm 1 \quad \tan \beta = \pm 2$$

Now $\tan \alpha$ & $\tan \beta$ are

positive since α and β

lie between 0 and $\pi/2$

$$\therefore \tan \alpha = 1 \text{ and } \tan \beta = 2$$

$$\therefore V = \pi(2-1) = \pi \text{ unit}^3$$

(3)

$$(iii) A_{48} = 20000(1.01)^{48} - M(1.01)^{48} - M(1.01)^{47} - \dots - M(1.01) - M$$

$$= 20000(1.01)^{48} - M(1.01)^{48} - M(1.01)^{47} - \dots - M(1.01) - M$$

$$= 20000(1.01)^{48} - M[1.01 + 1.01 + \dots + 1.01 + 1.01]$$

$$= 20000(1.01)^{48} - M[1.01 + 1.01 + \dots + 1.01 + 1.01]$$

a geometric series with

$$a = 1, r = 1.01, n = 48$$

$$= 20000(1.01)^{48} - M \left(\frac{1.01^{48} - 1}{1.01 - 1} \right)$$

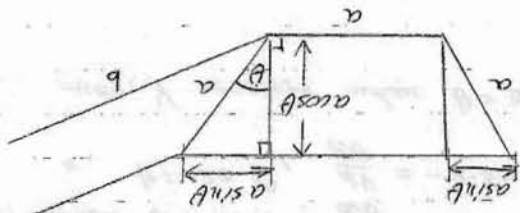
$$\therefore \text{Since } A_{48} = 0, \quad 20000(1.01)^{48} - M \left(\frac{1.01^{48} - 1}{0.01} \right) = 20000(1.01)^{48}$$

$$\therefore M \left[\frac{2(1.01)^{48}}{1.01 - 1} + \frac{1.01^{48} - 1}{0.01} \right] = 20000(1.01)^{48}$$

$$61.20680318 M = 31609.17709$$

$$M = 516.4324151$$

$$\therefore M = \$ 516.43$$



$$(i) A = \frac{1}{2}(2a + 2am \cos \theta) \times a \cos \theta$$

$$= (a + am \cos \theta) a \cos \theta$$

$$= a^2 \cos^2 \theta + a^2 m \cos^2 \theta$$

$$V = b(a^2 \cos^2 \theta + a^2 m \cos^2 \theta)$$

$$= a^2 b \cos^2 \theta + a^2 b m \cos^2 \theta$$

$$= a^2 b \cos^2 \theta (1 + m)$$

as required

(2)

$$\begin{aligned}
 \text{(ii)} \quad \frac{dV}{d\theta} &= a^2 b \times \frac{d}{d\theta}(\cos\theta) + a^2 b \times \frac{d}{d\theta}(\sin\theta \cos\theta) \\
 &= -a^2 b \sin\theta + a^2 b (-\sin^2\theta + \cos^2\theta) \\
 &= a^2 b (-\sin\theta - \sin^2\theta + \cos^2\theta) \\
 &= a^2 b (-\sin\theta - \sin^2\theta + 1 - \sin^2\theta) \\
 &= a^2 b (1 - 2\sin^2\theta - \sin\theta) \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \text{For maximum } V, \quad \frac{dV}{d\theta} &= 0. \\
 \therefore 1 - 2\sin^2\theta - \sin\theta &= 0 \quad \text{since } a^2 b > 0. \\
 2\sin^2\theta + \sin\theta - 1 &= 0. \\
 (2\sin\theta - 1)(\sin\theta + 1) &= 0. \\
 \therefore 2\sin\theta - 1 = 0 \quad \text{or} \quad \sin\theta + 1 = 0. \\
 \sin\theta &= \frac{1}{2} \quad \sin\theta = -1 \\
 \therefore \theta &= \pi/6 \quad \text{or} \quad \theta = -\pi/2, 3\pi/2, \dots
 \end{aligned}$$

$$\begin{aligned}
 \therefore \theta &= \frac{\pi}{6} \quad \text{since } \theta \text{ is acute} \\
 \therefore \theta &= 30^\circ.
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, when } \theta &= 29.9^\circ, \quad \frac{dV}{d\theta} = 4.5 \times 10^{-3} > 0 \\
 \text{" } \theta &= 30.1^\circ, \quad \frac{dV}{d\theta} = -4.5 \times 10^{-3} < 0
 \end{aligned}$$

$$\therefore \text{max. } V \text{ occurs when } \theta = 30^\circ.$$

(2)